### Q1. What is JavaScript? Mention its features and applications.

**JavaScript** is a scripting language used to create interactive and dynamic web pages. It is executed by a web browser and can be embedded within an HTML document using the `<script>` element.

**Features and applications:**

- Case-sensitive language.
- Supports objects, properties, and methods.
- Can process user input.
- Can perform calculations and logical operations.
- Can dynamically modify web page content.
- Supports loops and decision-making statements.
- Used with forms, events, HTML elements, and HTML5 Canvas.

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### Q2. Differentiate between Java and JavaScript.

| Java | JavaScript |
|---|---|
| A general-purpose programming language. | A scripting language primarily used in web applications. |
| Usually compiled to bytecode and executed using the JVM. | Traditionally executed by a JavaScript engine in the web browser. |
| Supports class-based object-oriented programming. | Supports object-based programming concepts. |
| Commonly used for standalone, enterprise, and other applications. | Mainly used to add dynamic behavior and interactivity to web pages. |
| Java programs are executed through a Java runtime environment. | JavaScript can be embedded in HTML and executed by the browser. |

**Conclusion:** Despite their similar names, Java and JavaScript are distinct languages with different execution models and purposes.

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### Q3. What are objects, properties, methods, and arguments in JavaScript?

#### Object

An **object** is an entity or thing that can contain properties and methods.

#### Property

A **property** is an attribute that describes an object.

**Example:**

```
car.color = "red";
```

Here, `car` is the object and `color` is its property.

#### Method

A **method** is an action or function associated with an object.

**Example:**

```
document.write("Hello");
```

Here, `document` is the object and `write()` is the method.

#### Argument

An **argument** is the value passed to a method or function.

In:

```
document.write("Hello");
```

`"Hello"` is the argument.

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### Q4. What are JavaScript variables? State the rules for naming variables.

A **variable** is a named memory location used to store data that may be used or modified during program execution.

#### Example:

```
var age = 25;
```

#### Rules for naming variables:

- A variable name may contain letters, digits, `_`, and `$`.

- It should not begin with a digit.
- It should not contain spaces.
- It should not use reserved keywords.
- JavaScript is case-sensitive; therefore, `Age` and `age` are different identifiers.

Variables are used to store user input, counters, numeric values, and other program data.

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### Q5. What is an array in JavaScript? Explain indexing.

An **array** is a data structure used to store multiple values under a single variable name. Individual elements are accessed using an **index or subscript** enclosed within square brackets `[]`.

**Example:**

```
var months = new Array();
months[1] = "January";
months[2] = "February";
```

An element can be accessed as:

```
document.write(months[1]);
```

The value inside square brackets identifies the specific element of the array.

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### Q6. What is a `while` loop? Explain with syntax.

A **`while` loop** is an iterative control statement that repeatedly executes a block of code as long as a specified condition evaluates to true.

**Syntax:**

```
while (condition) {
   // statements
}
```

**Example:**

```
var count = 1;

while (count <= 5) {
    document.write(count + "<br>");
    count = count + 1;
}
```

The loop terminates when the condition becomes false. Therefore, the loop-control variable must be updated properly to prevent an infinite loop.

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### Q7. What is an HTML Form? State the purpose of the `<form>` tag.

An **HTML form** is a mechanism for collecting information from users and submitting that information for processing.

The `<form>` tag acts as a container for form controls such as:

- Text fields
- Password fields
- Radio buttons
- Checkboxes
- Submit buttons
- Text areas
- Drop-down lists

**Basic syntax:**

```
<form action="process.php" method="post">
    <!-- form controls -->
</form>
```

The form element defines how and where the user-entered data will be submitted for processing.

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### Q8. Explain the `ACTION`, `METHOD`, and `ENCTYPE` attributes of the FORM element.

#### 1. `ACTION`

The `action` attribute specifies the destination or program that processes the submitted form data.

```
<form action="process.php">
```

**2. `METHOD`**

The `method` attribute specifies the HTTP method used to transmit form data. The common methods are:

- `GET`
- `POST`

```
<form method="post">
```

**3. `ENCTYPE`**

The `enctype` attribute specifies the encoding used when form data is submitted.

A commonly used encoding is:

```
enctype="application/x-www-form-urlencoded"
```

Thus, these attributes collectively control the **destination, transmission method, and encoding** of form data.

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### Q9. Differentiate between GET and POST methods.

| GET | POST |
|---|---|
| Form data is typically transmitted as part of the URL. | Form data is transmitted in the request body. |
| Data can appear as a query string. | Data is not placed in the URL query string in the same manner. |
| Commonly used for retrieval-oriented requests. | Commonly used for form submissions and data processing. |
| URL length limitations can affect the amount of transmitted data. | Better suited for transmitting larger amounts of data. |
| A submitted URL can be bookmarked or copied with its query parameters. | Request data is not represented as the same query string in the URL. |

**Example:**

```
<form method="get">
```

```
<form method="post">
```

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### Q10. Differentiate between radio buttons and checkboxes.

#### Radio Button

A **radio button** is used when the user must select **one option from a group**. Radio buttons belonging to the same group use the same `name` attribute.

```
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female
```

#### Checkbox

A **checkbox** is used when multiple options may be selected independently.

```
<input type="checkbox" name="sports" value="cricket"> Cricket
<input type="checkbox" name="sports" value="football"> Football
```

#### Difference

- **Radio button:** Usually allows one selection from a group.
- **Checkbox:** Allows independent and potentially multiple selections.

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### Q11. What is HTML5 Canvas? What are Immediate Mode and Retained Mode?

#### HTML5 Canvas

The **HTML5 Canvas** is a bitmap-based drawing area that can be manipulated using JavaScript and CSS. It is used for graphical and dynamic rendering.

Canvas properties include:

- `width`
- `height`
- `id`

#### Immediate Mode

Canvas follows an **immediate-mode rendering model**. The application redraws the required graphical output using Canvas API calls, particularly during animation or frame updates.

#### Retained Mode

In a **retained-mode model**, graphical objects are maintained by the rendering system, and changes to an object's properties can cause the displayed object to be updated.

#### Key difference

- **Immediate Mode:** The programmer is responsible for explicitly redrawing the scene.
- **Retained Mode:** The rendering system maintains and updates graphical objects.

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### Q12. What are the uses of HTML5 Canvas? Mention important Canvas drawing operations.

HTML5 Canvas is used to create dynamic graphical content in a web page.

#### Major uses:

- Drawing paths and shapes
- Drawing lines and arcs
- Creating rectangles and other graphics
- Displaying text
- Animation
- Mouse interaction
- Timed events
- Drawing video content
- Web-based games and interactive applications

#### Important drawing operations:

```
ctx.beginPath();
ctx.moveTo(x, y);
ctx.lineTo(x, y);
ctx.arc(x, y, radius, startAngle, endAngle);
ctx.fill();
ctx.stroke();
```

Other important operations include:

- `fillText()` — draws text
- `clearRect()` — clears a rectangular region
- `translate()` — changes the coordinate origin
- `scale()` — scales the coordinate system

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